

Flexing Innovation With PRINTED SENSORS In Modern Electronics

Traditional sensors have long held sway due to their established reliability and cost-efficiency. Yet, as technological demands advance, the distinctive attributes of printed sensors—namely, their flexibility and multifunctionality—take centre stage.

AKANKSHA SONDHI
GAUR

From their ability to seamlessly integrate into everyday objects like cars and healthcare devices, enhancing user experience with touch-sensitive controls and advanced security measures, to their role in pushing the boundaries of what is possible in electronics, sensors are more than just components—they are catalysts for change. Let's dive into their versatility, the latest breakthroughs in materials, the hurdles they face in widespread adoption, and the exciting possibilities they hold for shaping the future of technology.

Sensors play a crucial role in blending our physical surroundings with digital capabilities, fundamentally altering how we interact with our devices. Printed and flexible sensors distinguish themselves through innovative deposition techniques on materials like plastics and silicones. They excel in scalability, covering expansive areas

while maintaining sleek profiles, and offering multifaceted functionalities such as temperature and pressure monitoring. Their adaptability extends to detecting diverse parameters like force, touch, light, gases, and biometrics. For instance, a pressure sensor might utilise conductive carbon ink to measure force accurately. These cutting-edge sensors are poised to revolutionise large surfaces, potentially reimagining environments such as automotive interiors with seamlessly integrated touch and heating functionalities.

Driving growth: Material science for sensors

Material science advancements are pivotal in enhancing the capabilities of printed sensors. By developing new materials with improved conductive properties, sensitivity, and durability, researchers can create sensors that operate more efficiently and reliably under various conditions. These advancements enable the

detection of a broader range of stimuli, crucial in complex IoT systems where multiple environmental factors need monitoring simultaneously.

Printed and flexible sensors use a variety of materials to maximise performance. Epoxy and silicones provide adhesion and act as barrier layers, complemented by metals, polymers such as polyimide, polyethylene naphthalate, polyethylene terephthalate,

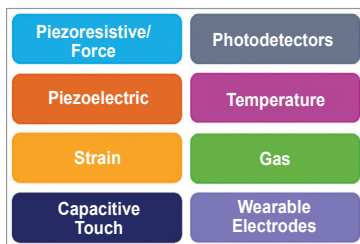


Fig. 1: Printed sensor technologies

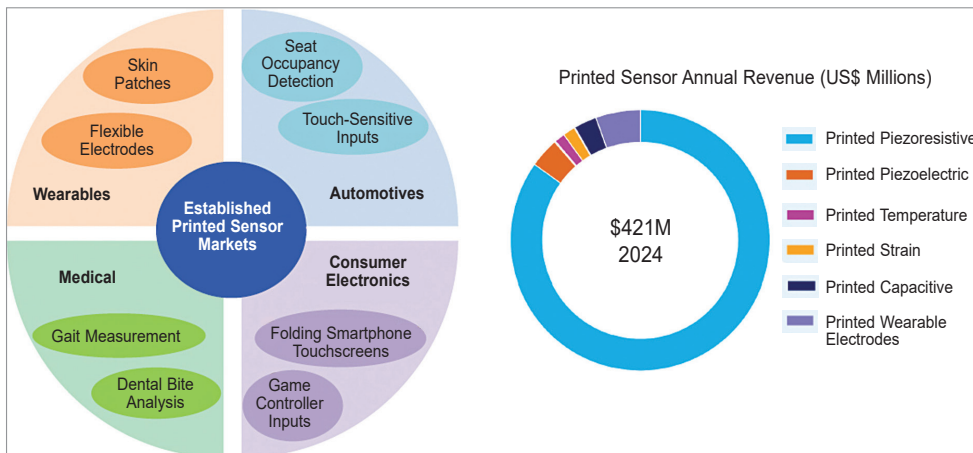


Fig. 2: Market successes for printed and flexible sensors